



Master Presenter's Manual

for

Programming in C

Curriculum Code: OV-1503

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Ver. 1.0***

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	February 2024	New	-

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1. Introduction

At the end of this course, students will be able to:

- Describe parts of a computer
- List new technologies in computer
- Explain data storage in computers
- Describe the evolution of computer languages
- List the different modeling tools for program designing
- Explain algorithm
- Explain flowcharts and its symbols
- Explain pseudocodes
- Explain IF and IF..ELSE selection construct
- Explain case construct
- Define operators
- List the different types of operators
- Describe the use of arithmetical and relational operators
- Identify the precedence of operators in an expression
- Explain looping constructs
- Describe the different types of loops
- Define structured programming and its benefits
- Identify the modeling tool used for structured programming
- Describe the top-down approach in program designing
- Define arrays
- List the different types of arrays
- Describe the searching and sorting techniques
- Explain file organization
- Describe different types of files
- Explain the different file operations
- Describe system flowcharts
- Explain DFDs and decision tables
- Describe HIPO charts
- Define software programs
- Describe the program development process
- Explain programming languages
- Explain program execution and program testing
- Explain the basic concept of program, including variable, data types and expressions
- Explain the basic Flow Control of a C program
- Explain point of C program focusing data and structured programming
- Explain array, structures, pointers, basic arithmetic and other basic data structures

2. Information on Module Allocation

Module	Online (No. of Hrs)	Skill Code
C	32	75

Module Deliverables available on OnlineVarsity

To aid the learning process, following are the deliverables

Student Deliverables:

1. Learner's Guide (eBook)

Resources available on OnlineVarsity for Students:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Glossary - Student can access a list of subject related specialized words with their definitions.
	FAQ - Student can access frequently asked questions and their answers.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

3. Week-wise Session Schedule

- A session is of 2 hours duration.

Week	Day 1	Day 2	Day 3	Day 4
1	Session 1 C-TL1	Session 2 C-TL2	Session 3 C-TL3	Session 4 C-TL4
2	Session 5 C-TL5	Session 6 C-TL6	Session 7 C-TL7	Session 8 C-TL8
3	Session 9 C-TL9	Session 10 C-TL10	Session 11 C-TL11	Session 12 C-TL12
4	Session 13 C-TL13	Session 14 C-TL14	Session 15 C-TL15	Session 16 C-TL16

C: Logic Building and Elementary Programming
TL: Online Session

4. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	C – TL1	All the topics as listed below from Session 1 of <i>Elementary Programming in C</i> book should be covered in this session. <u>Session 1: Basics of C</u> • Introduction to programs • Constituents of any programming language • Algorithms, pseudo codes and flowcharts • Introduction to C	Elementary Programming in C SG – Session 1 XP – Session 1 TG – Session 1
2	C – TL2	All the topics as listed below from Session 2 of <i>Elementary Programming in C</i> book should be covered in this session. <u>Session 2: Variables and Data Types</u> • C language specific data storage mechanisms • Data Types • Variables and constants in C • Basic arithmetic operators with the variables and constants	Elementary Programming in C SG –Session 2 XP –Session 2 TG –Session 2
3	C – TL3	All the topics as listed below from Session 4 of <i>Elementary Programming in C</i> book should be covered in this session. <u>Session 4: Operators and Expressions</u> • Assignment Operators • Relational Operators • Logical operators • Expressions • Operator precedence	Elementary Programming in C SG – Session 4 XP – Session 4 TG – Session 4
4	C – TL4	All the topics as listed below from Session 6 of <i>Elementary Programming in C</i> book should be covered in this session.	Elementary Programming in C SG – Session 6 XP – Session 6

Session No.	Session Title	Session Details	Deliverables' Mapping
		<u>Session 6: Input and Output in 'C'</u> • Introduction to Input Output in C • printf(), scanf(), getchar(), and putchar()	TG – Session 6
5	C – TL5	All the topics as listed below from Session 7 of <i>Elementary Programming in C</i> book should be covered in this session. <u>Session 7: Condition</u> • Selection constructs • Using if • Nested ifs • if-else, • multiple ifs • switch-case <u>Session 9: Loop</u> • Introduction to loops • for • while • do-while • break • continue • exit()	Elementary Programming in C SG – Session 7 & 9 XP – Session 7 & 9 TG – Session 7 & 9
6	C – TL6	All the Lab topics from Session 3, Session 5, and Session 8 of <i>Elementary Programming in C</i> book should be covered in this session.	Elementary Programming in C Session 3, 5, & 8
7	C – TL7	All the Lab topics from Session 10 of <i>Elementary Programming in C</i> book should be covered in this session.	Elementary Programming in C Session 10
8	C – TL8	All the topics as listed below from Session 11 of <i>Elementary Programming in C</i> book should be covered in this session. <u>Session 11: Arrays</u> • Introduction to the array data structure	Elementary Programming in C SG – Session 11 XP – Session 11 TG – Session 11

Session No.	Session Title	Session Details	Deliverables' Mapping
		- Defining and using arrays - Creating and initializing String/Character arrays - Multidimensional arrays	
9	C – TL9	All the topics as listed below from Session 13 of <i>Elementary Programming in C</i> book should be covered in this session. <u>Session 13: Pointers</u> - Introduction to pointers - Using pointers - Comparison of pointers - Pointer arithmetic - Passing pointers as arguments to functions - Memory allocation - Relation of pointers with character arrays and strings	Elementary Programming in C SG – Session 13 XP – Session 13 TG – Session 13
10	C – TL10	All the Lab topics from Session 12 and Session 14 of <i>Elementary Programming in C</i> book should be covered in this session.	Elementary Programming in C Session 14
11	C – TL11	All the topics as listed below from Session 15 of <i>Elementary Programming in C</i> book should be covered in this session <u>Session 15: Functions</u> - User defined functions in C - Structure of functions - Declaration of functions and their prototypes - Calling functions - Call by Value - Call by Reference - Scope of variables (global and formal parameters) - Storage classes and function pointers	Elementary Programming in C SG – Session 15 XP – Session 15 TG – Session 15
12	C – TL12	All the topics as listed below from Session 17 of <i>Elementary Programming in C</i> book should be covered in this session	Elementary Programming in C SG – Session 17

Session No.	Session Title	Session Details	Deliverables' Mapping
		<u>Session 17: Strings</u> • Concept of strings in C • Pointers to strings • String input and output functions • Various string functions • Passing arrays as function arguments • Passing strings as function arguments	XP – Session 17 TG – Session 17
13	C – TL13	All the Lab topics from Session 16 and Session 18 of <i>Elementary Programming in C</i> book should be covered in this session.	Elementary Programming in C Session 18
14	C – TL14	All the topics as listed below from Session 19 of <i>Elementary Programming in C</i> book should be covered in this session. <u>Session 19: Advanced Data types and Sorting</u> • Need for structures • Defining structures • Assignment of values to structure elements • Accessing structure elements • Passing structures as function arguments • Arrays of structures • Pointers to structures • Passing pointers to structures as function arguments • Use of 'typedef' • 'Insertion Sort' and 'Bubble Sort' algorithms	Elementary Programming in C SG – Session 19 XP – Session 19 TG – Session 19
15	C – TL15	All the topics as listed below from Session 21 of <i>Elementary Programming in C</i> book should be covered in this session. <u>Session 21 - File Handling</u> • Introduction to Streams • Text Streams and Binary Streams • File Pointers and Various File	Elementary Programming in C SG – Session 21 XP – Session 21 TG – Session 21

Session No.	Session Title	Session Details	Deliverables' Mapping
		Functions • Current File Pointer • Command-Line Arguments	
16	C – TL16	All the Lab topics from Session 20 and Session 22 of <i>Elementary Programming in C</i> book should be covered in this session.	Elementary Programming in C Session 22

5. Deliverables

The deliverables to be provided to the students and the faculty have been tabulated below:

➤ Deliverables for Student:

Module	Deliverable
Logic Building and Elementary Programming	Kit Code: OV-1503-KIT01-01-A Kit Name: Kit of 1 title for OV-1503 Programming in C

Deliverables for Faculty:

Module	Deliverable
Logic Building and Elementary Programming	Kit Code: OV-1503-KIT01-01-A Kit Name: Kit of 1 title for OV-1503 Programming in C
Course	Classroom Presentations Trainer Guide Master PM

6. Library References

➤ Principles and Techniques of Programming , by T.M.Ramchandran
➤ Programming Logic and Design, Comprehensive by Joyce Farrell

7. Evaluation Strategy

The Evaluation Strategy for this course is as follows:

- Final Examination

7.1 Award of Grades

Overall Weighted Marks $\geq 40\%$ but $< 60\%$ qualifies for **PASS**

Overall Weighted Marks $\geq 60\%$ but $< 75\%$ qualifies for **CREDIT and**

Overall Weighted Marks $\geq 75\%$ qualifies for **DISTINCTION**

Note: To attain a PASS/CREDIT/DISTINCTION grade, a student should achieve at least 40% in Final Examination; otherwise he/she will be declared as 'Referred'.

7.2 Final Examination

Below is the list of modules to be covered in Final Examination along with their weightages:

Final Examination (E1)

Marks: 50

Duration: 60 minutes

Type: Objective

Sr. No.	Name of the Subject	Distribution of Marks
1.	Logic Building and Elementary Programming	50

Final Examination (R1)

Marks: 50

Duration: 120 minutes

Type: Practical

Sr. No.	Name of the Subject	Distribution of Marks
1.	Logic Building and Elementary Programming	50

Evaluation Details

Practical – 50%

OES – 50%

8. Classroom and Lab Pre-requisites

The requirements for conducting the module of this course have been described as follows:

8.1 Software Requirements

Microsoft Windows 10
NetBeans IDE latest version

8.2 Recommended Hardware

Intel Core i3/i5 Processor or higher
8 GB RAM or above
Color SVGA
500 GB Hard Disk space
Mouse
Keyboard

8.3 Other Requirements (For Classroom)

Computer for giving demonstrations wherever applicable
LCD projector

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